Propidium Iodide (PI) staining

- **Intact cell membrane**: Propidium iodide (PI) cannot readily penetrate an intact cell membrane. Therefore, PI fluorescence remains outside the cell, allowing the cell boundary to be identified.
- **Damaged cell membrane**: When the cell membrane is damaged, PI can enter the cell and bind to nucleic acids in the nucleus, resulting in strong, localized PI signals that appear as bright blobs.
- **Prolonged treatment**: Extended exposure to PI may also lead to PI uptake by cells with initially intact membranes, potentially resulting in intracellular PI signals.

Propidium Iodide (Sigma-Aldrich, Cas No. P4170)

Chemical	M.W. (g/mol)	Stock conc.	Working conc.	Location
Propidium iodide	668.39	1.5 mM	15 μM	4°C (a)

- **Stock solution (1.5 mM)**: Dissolve **1 mg / mL** in ddH_2O, stored at **4°C**, **protected from light**.
- **Working solution (15 μM)**: Dilute 10 µL stock in 1 mL ddH_2O (so the working concentration is 10 µg / mL), stored at **4°C**, **protected from light**.
- The stock (1 mg / mL) and working solution (10 µg / mL) are also located at 4°C (a).

Confocal parameter

Parameters	Theoretical value	Our machine
Excitation peak	493 nm	506 nm
Emission peak	636 nm	530 ~ 700 nm
Magnification		10X
Laser intensity		16.0%
Master Gain		700 V
Digital Gain		1.0
Pinhole		≈ 32 μm
Z-stack interval		4 μm
Scan speed		6
Scan direction		↔

> Not binding to DNA → Excitation / Emision maxima: 493 / 636 nm
> After binding to DNA → Excitation / Emission maxima: 535 / 617 nm

Example image

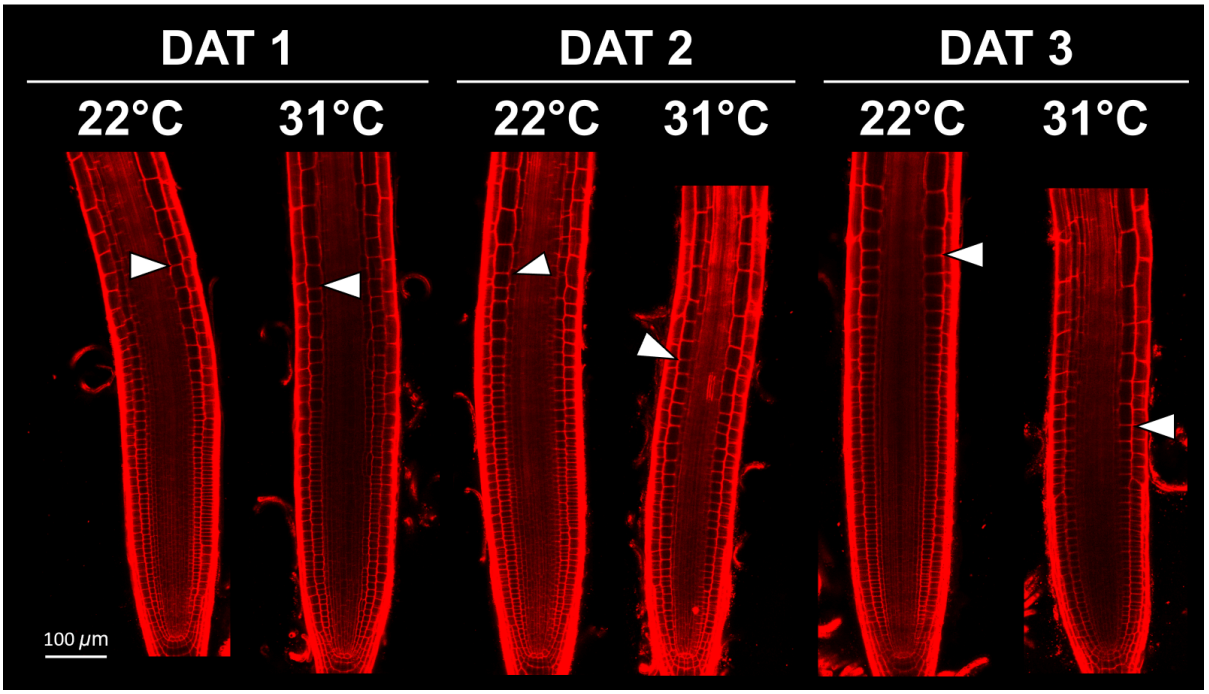


Figure 1. Confocal laser scanning microscopy images of *Arabidopsis thaliana* roots stained with propidium iodide.

If you encounter any issues with the protocol, please report them at: https://github.com/P10911004-NPUST/jklai/issues.